

FULL-STACK WEB DEVELOPER - MERN STACK (FSWDMEN)

syllabus

FULL-STACK WEB DEVELOPER -MERN STACK

DURATION: 216 HOURS/6 MONTHS

OBJECTIVE

The Objective of this course is to learn the core concepts of both the frontend and backend programming, to get familiar with the latest web development technologies like Mongo DB, Express Js, React JS, Node Js, learn all about Non-SQL databases learn complete web development process.

EXIT PROFILE

- ✓ Complete Knowledge in Web Designing& Development
- ✓ Front -End Development
- ✓ Back End-Development
- ✓ Database Programming

CAREER PATH

- ✓ Full-Stack Web Developer - MERN
- ✓ Frontend developer
- ✓ Back-End Developer
- ✓ UI Designer
- ✓ Non-SQL Database Programmer

FACULTY SKILL SET

- ✓ UI Designing
- ✓ HTML, CSS
- ✓ JavaScript
- ✓ JQuery
- ✓ Knowledge in Programming

HARDWARE AND SOFTWARE REQUIREMENTS

- ✓ Operating System: Windows 8/10/11
- ✓ Minimum Memory: 4 GB
- ✓ Recommended Memory: 4 GB / 8 GB
- ✓ Minimum Disk Space: 160 GB
- ✓ Recommended Disk Space: 500 GB
- ✓ Software: Mongo DB, Express Js, React JS, Node Js

PREREQUISITE FOR STUDENTS

Before attending this course, students must have:

- ✓ Basic Knowledge of HTML, CSS
- ✓ Knowledge in web programming will be added advantage

COURSE OUTLINE			
COURSE NAME:	FULL STACK WEB DEVELOPER -MERN STACK	DURATION:216 H/6 M	
MODULE	TOPICS	DURATION	TOTAL
Module - I	Front end development: UI designing	6H/4D	90H/2.5 M
	HTML	12H/8D	
	CSS	12H/8D	
	JavaScript	30H/20D	
	JQuery	18H/12D	
	Bootstrap	12H/8D	
Module - II	Mongo DB	27H/18D	126H /3.5 M
	Express JS	27H/18D	
	Node JS	27H/18D	
	React JS	27H/18D	
	Working with Server, Various APIs.	6H/4D	
	Project Work	12H/8D	

MODULE IN DETAIL

FRONT-END DEVELOPMENT

UI DESIGNING

- ✓ Colour Concepts
- ✓ Menus
- ✓ Tools
- ✓ Colour Theory
- ✓ Typography
- ✓ Introduction to UI design
- ✓ Real-time designing of a Professional Website.

HTML

- ✓ introduction
- ✓ Bulding Blocks of Web Frontend
- ✓ Introduction To HTML
- ✓ Basic HTML TAGS
- ✓ Advanced HTML Tags & concepts

CSS

- ✓ Introduction to CSS
  - What is CSS?
  - CSS Syntax
  - Three ways to write CSS
  - CSS Priorities
  - Cascading Effect of CSS Properties

- ✓ CSS Basics
  - CSS Selectors
  - Basic CSS Properties (Color, Background, Padding, Margin, Width, Height, Font properties etc)
  - Box Model
  - Units and Colors (px % etc. named colors Hex color representation etc)
  - Using Browser tools to Check CSS Properties Rendered.
- ✓ CSS Advanced
  - Max and min widths, Inline Block, z-index, align etc
  - Pseudo Classes
  - Box Sizing, Display, Position etc
  - CSS Layouts
  - Flex and Grid
  - CSS Transformations
  - CSS Animations
  - Responsive Design (Media Queries)

JAVASCRIPT

- ✓ Introduction to DHTML
  - What is DHTML
  - History of Different ways dynamic elements were added to HTML Pages
- ✓ Javascript Basics
  - Writing JavaScript in Browser Console
  - Writing JavaScript in Web pages
  - Linking External
  - Using Console.log to debug the script ins browser Console.
- ✓ Javascript Advanced
  - Variables, operators, datatypes, loops, Conditions
  - Objects and Arrays
  - Accessing Arrays and Objects
  - Functions
  - Functions within Objects
  - JSON
- ✓ Document Object Model
  - What is DOM
  - Viewing the DOM in Browser Developer Tools
- ✓ Manipulating DOM using JS
  - Accessing document object
  - Accessing elements using ID, Class and tag
  - Manipulating existing elements
  - Creating and adding New Elements to document object
- ✓ DOM Events
  - What are events
  - Different types of events
  - Call a function on onclick event

- Adding event listeners
- Form validation Using JS

✓ Project - Create a To Do List website prototype using HTML, CSS and JS

JQUERY

✓ JQuery Basics
- What is JQuery?
- Adding JQuery to a web page
- JQuery Syntax
- JQuery Selector
- JQuery Events

✓ JQuery Advanced
- HTML Manipulations
- Effects
- Traversing
- Plugins

BOOTSRAP

✓ What is Bootstrap?
✓ Adding bootstrap to your projects.
✓ Different predefined Element styles
✓ The Grid System
✓ Forms
✓ Templates

BACK-END DEVELOPMENT USING PYTHON AND ITS FRAMEWORKS

MONGO DB

INTRODUCTION TO NOSQL

✓ Introduction
✓ Advantages of NoSQL
✓ Disadvantages of NoSQL
✓ Types of NoSQL database
✓ Strengths and weaknesses of NoSQL

INTRODUCTION TO MONGODB

✓ Overview
✓ Advantages of MongoDB over RDBMS
✓ Features of MongoDB
✓ Applications of MongoDB
✓ Installation and Environment Setup

DATA MODELLING& BASIC OPERATIONS

✓ Introduction
✓ Data Model Design
✓ Considerations while designing Schema in MongoDB
✓ Create Database

- ✓ Drop Database
- ✓ MongoDB Datatypes

COLLECTIONS AND BASIC OPERATIONS ON COLLECTIONS

- ✓ Introduction
- ✓ Creating a Collection
- ✓ Dropping Collections

INSERTING DATA INTO TABLES

- ✓ The insert () Method
- ✓ The insertOne () method
- ✓ The insertMany () method

QUERY DOCUMENT

- ✓ The find () Method
- ✓ The pretty () Method
- ✓ The findOne() method
- ✓ RDBMS Where Clause Equivalents in MongoDB
- ✓ AND in MongoDB
- ✓ OR in MongoDB
- ✓ NOR in MongoDB
- ✓ NOT in MongoDB

UPDATE DOCUMENT

- ✓ MongoDB Update () Method
- ✓ MongoDB Save () Method
- ✓ MongoDB findOneAndUpdate () method
- ✓ MongoDB updateOne () method
- ✓ MongoDB updateMany () method

DELETE DOCUMENT

- ✓ The remove () Method
- ✓ Remove Only One RECORD
- ✓ Remove All Documents
- ✓ LIMIT RECORDS
- ✓ The Limit () Method
- ✓ MongoDB Skip () Method
- ✓ Sorting the Data
- ✓ Indexing the Data
- ✓ Aggregate Functions
- ✓ Data Replication

SHARDING

- ✓ Introduction
- ✓ Requirement for Shading
- ✓ Sharding in MongoDB

BACKUP AND RESTORE

- ✓ Dump MongoDB Data
- ✓ Restore data

DEPLOYMENT

- ✓ Introduction
- ✓ mongostat Command
- ✓ mongotop Command

ADVANCED FEATURES OF MONGODB

- ✓ Relationships
- ✓ Database References
- ✓ Covered Queries
- ✓ Analyzing Queries
- ✓ Atomic Operations
- ✓ Advanced Indexing
- ✓ Map Reduce
- ✓ Grids
- ✓ Capped Collections

EXPRESS JS

OVERVIEW

- ✓ Definition
- ✓ Pug
- ✓ Mongoose

ENVIRONMENT

- ✓ NPM
- ✓ Installation using NPM
- ✓ Setting up an application
- ✓ app.get(route, callback)
- ✓ res.send()
- ✓ app.listen(port, [host], [backlog], [callback]])

ROUTING

- ✓ app.method(path, handler)
- ✓ Routers

HTTP METHODS

- ✓ GET
- ✓ POST
- ✓ PUT
- ✓ DELETE

URL BUILDING

- ✓ Introdcution
- ✓ Pattern Matched Routes

MIDDLEWARE

- ✓ Introdcution
- ✓ Order of Middleware Calls
- ✓ Third Party Middleware

TEMPLATING

- ✓ Introduction to PUG
- ✓ Features of PUG
- ✓ Passing Values to Templates
- ✓ Conditionals
- ✓ Include and Components

STATIC FILES

- ✓ Serving static files
- ✓ Multiple Static Directories
- ✓ Virtual Path Prefix

FORM DATA

- ✓ Introdcution
- ✓ Working with Forms

DATABASE

- ✓ Introdcution to Mongoose
- ✓ Setting up Mongoose
- ✓ Saving Documents
- ✓ Retrieving Documents
- ✓ Updating Documents
- ✓ Deleting Documents

COOKIES

- ✓ Introduction
- ✓ Adding cookies
- ✓ Adding Cookies with Expiration Time
- ✓ Deleting Existing Cookies

SESSIONS

- ✓ Introdcution
- ✓ Installation and Setup of Express Session

AUTHENTICATION

- ✓ Introdcution
- ✓ SIGNUP.JADE

RESTFUL APIS

- ✓ Introdcution
- ✓ URIs and Methods
- ✓ GET routes
- ✓ POST route
- ✓ PUT route
- ✓ DELETE route

SCAFFOLDING

- ✓ Introdcution
- ✓ Yeoman.
- ✓ Installing and setting up Yeoman.

ERROR HANDLING

- ✓ Introdcution
- ✓ Methods of Error Handling

DEBUGGING

- ✓ Introdcution
- ✓ Debugging Techniques in ExpressJS

REACT JS

INTRODUCTION

- ✓ Overview
- ✓ Versions
- ✓ Applications

INSTALLATION

- ✓ Prerequisite
- ✓ Toolchain
- ✓ The serve static server
- ✓ Babel compiler
- ✓ Create React App toolchain
- ✓ Advantages of using React toolchain

FEATURES

- ✓ Virtual DOM
- ✓ Components
- ✓ JSX

- ✓ One-way data binding
- ✓ Scalable
- ✓ Flexible
- ✓ Modular

ADVANTAGES & DISADVANTAGES

- ✓ Advantages of ReactJS
- ✓ Disadvantages of React

ARCHITECTURE

- ✓ Overview
- ✓ Architecture of the React Application
- ✓ Workflow of a React application

CREATING A REACT APPLICATION

- ✓ Rollup bundler
- ✓ Parcel bundler

JSX

- ✓ Using JSX in ReactJS
- ✓ Features of JSX
- ✓ Expressions in JSX
- ✓ Functions in JSX
- ✓ Attributes in JSX
- ✓ Using Expression within Attributes
- ✓ Nested Elements in JSX

COMPONENTS

- ✓ Overview
- ✓ Function Components
- ✓ Class Components
- ✓ Creating a React component
- ✓ Creating a class component
- ✓ Creating a function component
- ✓ Splitting Components
- ✓ Nested Components
- ✓ Using Components
- ✓ Component Collection

STYLING

- ✓ CSS stylesheet
- ✓ Inline styling
- ✓ CSS Modules

PROPERTIES (PROPS)

- ✓ Using Props
- ✓ Default Props
- ✓ State vs Props
- ✓ Creating Components Using Properties
- ✓ Objects as properties
- ✓ Props Validation
- ✓ Constructor

COMPONENT LIFE CYCLE

- ✓ Life Cycle API

EVENT MANAGEMENT

- ✓ Event Management in ReactJS
- ✓ Synthetic React Events
- ✓ Adding an Event
- ✓ Handling an Event
- ✓ Passing Arguments to Event Handler
- ✓ Creating an Event–Aware Component
- ✓ Passing Extra Information to Event Handler

STATE MANAGEMENT

- ✓ Overview
- ✓ Defining a State
- ✓ Creating a state Object
- ✓ STATE MANAGEMENT API
- ✓ Special Feature of React State API
- ✓ Stateless Component
- ✓ State Management Using React Hooks

PAGINATION

- ✓ Pagination component

MATERIAL UI

- ✓ Installation
- ✓ Examples

HTTP CLIENT PROGRAMMING

- ✓ Overview
- ✓ Expense Rest API Server

FORM PROGRAMMING

- ✓ Controlled Component
- ✓ Uncontrolled Component

- ✓ Formik

CONDITIONAL RENDERING

- ✓ Conditional rendering in React
- ✓ Methods of conditional rendering

LISTS & KEYS

- ✓ List and For
- ✓ List and keys

ROUTING

- ✓ React Router
- ✓ Install React Router
- ✓ Nested Routing
- ✓ Advantages of React Router

REDUX

- ✓ Overview
- ✓ Concepts
- ✓ Redux API
- ✓ Provider component

ANIMATION & BOOTSTRAP

- ✓ Transition
- ✓ CSS Transition
- ✓ Transition Group
- ✓ Integrating BOOTSTRAP

MAPS & TABLES

- ✓ Overview of MAPS
- ✓ Table component

MANAGING STATE USING FLUX

- ✓ Overview
- ✓ Applying flux

TESTING, BUILDING AND DEPLOYMENT

- ✓ Testing
- ✓ Building
- ✓ Deployment

PROJECT WORK

NODE JS

INTRODUCTION

- ✓ Overview
- ✓ Features
- ✓ Applications of Node JS

ENVIRONMENT SETUP

- ✓ Local Environment Setup
- ✓ Installation on Windows
- ✓ Installation on Ubuntu Linux

STARTING APPLICATIONS

- ✓ Console Application
- ✓ Creating Node.js Application

REPL TERMINAL

- ✓ Tasks
- ✓ Multiline Expression
- ✓ Underscore Variable
- ✓ Dot commands

COMMAND LINE OPTIONS

- ✓ Display version
- ✓ Evaluate script
- ✓ Start REPL
- ✓ Loading module

PACKAGE MANAGER (NPM)

- ✓ Overview
- ✓ Install Package Locally
- ✓ Install Package Globally
- ✓ Update Package
- ✓ Uninstall Packages

CALLBACKS CONCEPT

- ✓ Overview
- ✓ Blocking Code Example
- ✓ Non-Blocking Code Example
- ✓ Callback as Arrow function

UPLOAD FILES

- ✓ Overview
- ✓ Formidable

- ✓ Multer

EVENTS & EVENT LOOPS

- ✓ Overview
- ✓ Examples
- ✓ Event Loop

EVENT EMITTER & DEBUGGER

- ✓ Overview
- ✓ Examples
- ✓ Debugger

GLOBAL OBJECTS

- ✓ Buffer class
- ✓ Console class
- ✓ Process object
- ✓ Global timer functions
- ✓ Global variables
- ✓ Global Objects

CONSOLE

- ✓ Console Class
- ✓ Console Methods

PROCESS

- ✓ Process Events
- ✓ Process Methods
- ✓ Process Properties

SCALING APPLICATION

- ✓ Overview
- ✓ The exec() method
- ✓ The spawn() Method
- ✓ The fork() Method
- ✓ execFile() method

PACKAGING

- ✓ Nexe
- ✓ Features of Nexe
- ✓ Installation
- ✓ pkg tool
- ✓ JXCore

STREAMS

- ✓ Overview
- ✓ Types of Streams
- ✓ Piping the Streams
- ✓ Chaining the Streams

FILE SYSTEMS

- ✓ Synchronous vs Asynchronous Methods
- ✓ File Operations
- ✓ Directory Operations

PROJECT WORK